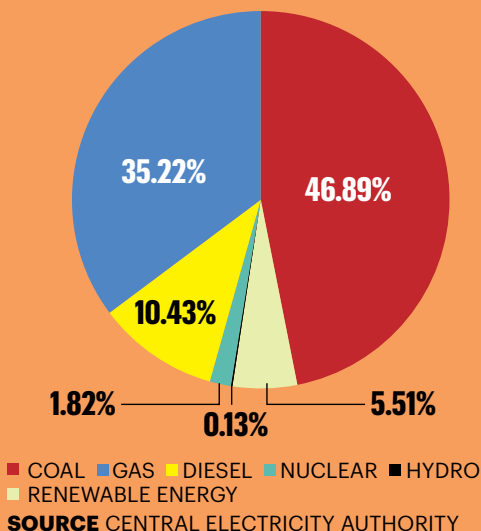


INSTALLED CAPACITY RESOURCE-WISE

Coal has the largest share in India's energy mix. Nuclear power is less than 2%



power capacity in India. Adani Group Chairman Gautam Adani recently visited Tarapur Atomic Power Station. The company is reportedly looking at a 30-GW nuclear power capacity in the long term. Reliance Industries Chairman Mukesh Ambani has also announced that the company will invest in nuclear power.

Aditya Birla-led Hindalco is exploring the sites for setting up captive nuclear power to decarbonise its aluminium and copper manufacturing operations. State-run power producer NTPC Ltd is also going big with a target of adding 30GW nuclear power by 2034. It recently invited tenders seeking global partners to establish large nuclear reactors with a combined capacity of 15GW.

Some of the renewables players were also part of the discussion as they are exploring nuclear power for integration with renewables to meet peak power demand. ACME Solar Holdings has also announced its interest in nuclear power. The Indian Railways is also venturing into the sector.

Globally, data centre players are also exploring captive nuclear power to fuel their energy-intensive operations through clean sources. Google, for instance, is collaborating with utilities in the US and other nations to explore the use of nuclear for powering its data centres, citing the need for clean and reliable power to drive artificial intelligence efforts. Amazon, too, has spent over \$500 million on nuclear technology, including the building of small modular reactors in partnership with companies like Dominion Energy, to generate at least 300 Mw in Virginia. Microsoft, too, has signed a significant agreement with Constellation Energy to reopen the Three Mile Island nuclear plant, purchasing carbon-free electricity for its data centres for the next 20 years.

COST COMPETITIVENESS

Going by the official numbers, the cost of building a BSR is around ₹15 crore per MW, depending on the mode of financing. A Parliamentary Standing Committee, in its latest report, acknowledged the high upfront costs and recommended a robust financial model that would include government incentives, VGF and sovereign guarantees to attract

WHAT IS BHARAT SMALL REACTOR

Bharat Small Reactors (BSRs) are compact nuclear reactors designed to produce electricity on a smaller scale than traditional sizeable nuclear power plants. These could be valuable for providing power to remote regions or acting as a dedicated energy source for large industries.

- 1 **220 MW capacity Pressurised Heavy Water Reactor type design**
- 2 **Technology designed and developed by NPCIL and BARC**
- 3 **Uranium to be provided by the government**
- 4 **Estimated timeline for construction of the first unit is 42-48 months**
- 5 **Cost estimated to be ₹15 crore per MW**